

MANUFACTURING UNDER PRESSURE

How the scientific and industrial community came together to address the demands of the healthcare system in a time of crisis

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As soon as the first case was identified in India in January last year, it was clear that the country would have to prepare quickly for the pandemic. All the three wings of the armed forces began to brace for the contagion, preparing quarantine and isolation centres, and the government began procurement of PPE kits and n95 masks. By the time the nationwide lockdown was imposed on 25 March, the contagion had raged through countries with the most advanced healthcare systems in the world. Apart from a shortage of healthcare workers and beds, there was a shortage of medical equipment, sanitation equipment, masks, sanitisers and testing kits. On 09 April 2020, the government exempted some much needed supplies from import cess and duties: Ventilators, face masks, surgical Masks, PPE kits, testing kits, and anything needed for the local production and manufacture of all of these. Within months, this demand was met.

VENTILATORS

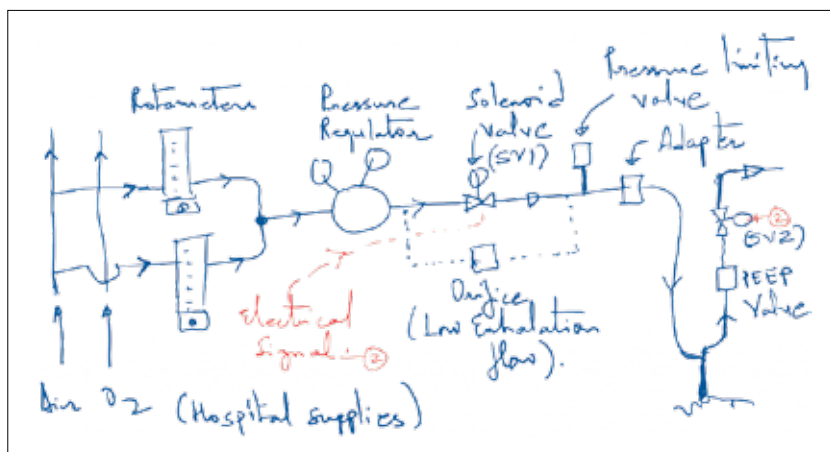
One of the medical equipment that medical professionals anticipated a shortage of was ventilators. Ventilators are devices that help the lungs breathe, and can either be invasive or non-invasive. The former involves a tube that goes straight down the windpipe from the mouth, into the lungs, while the latter uses a face

mask that seals around the nose and mouth of the patient, either of these are the interface for the patient. Oxygen and regular air are mixed together in a ventilator, which has electronic and pneumatic components. The ventilators can use either a compressor or a turbine to drive the air. While compressors need cylinders or a centralised supply, turbines work off the ambient air. A ventilator may use one of these two approaches. For use in the ICU the ventilator must be capable of delivering the air in two modes, which is a decision that the healthcare professional takes at the time of ventilation. These are volume controlled ventilation and pressure controlled ventilation, which means that a fixed amount of volume or pressure has to be delivered per tic, known as a ventilating cycle. This is what it means when a model is referred to as an "ICU ventilator" or

"Critical Care Ventilator" (CCV). On 30 March 2020, a DRDO technical committee came up with a list of essential features for a ventilator to combat the pandemic, These were:

1. The machine should be turbine/compressor based because the installation sites might not have central Oxygen Lines.
2. The machine should have Invasive, non-invasive and CPAP features to make them versatile.
3. 200-600 ML tidal volume, Lung Mechanics Display.
4. Monitoring of Plateau Pressure, PEEP, PS, oxygen concentration, lung mechanics/inverse ratio (I:E)
5. Pressure & volume control & PSV
6. Continuous working capability for 4-5 days

The important thing here is that the ventilator should very carefully accentuate the breathing of a person's damaged lungs, and too much pressure



Circuit plan of the IISC ventilator

Credit: IISC